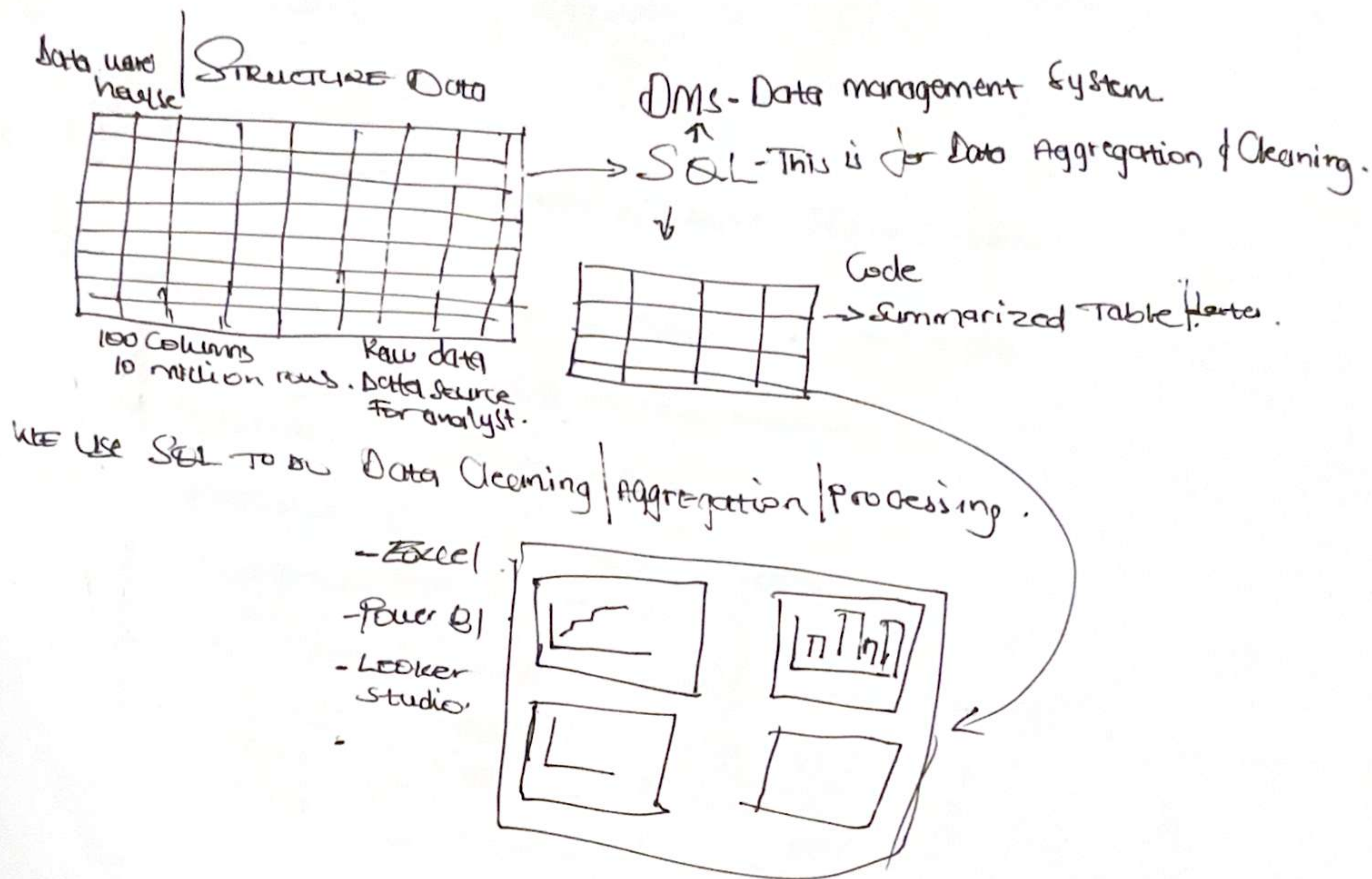


MONDAY - 09 FEBRUARY 2020.

SQL - STRUCTURED QUERY LANGUAGE.

INTRODUCTION TO SQL.



SQL Syntax means the way we write SQL.

- Syntax is the way you

1. SELECT Statement: allows us to Specify the Columns that we want from a specific table.

2. FROM: Allows you to Specify the Table you are selecting from.

. You Specify the Name of the Database, Schema, Table name.

SELECT
Column 1
Column 2
Column 3

FROM database. Schema. table_name.

e.g. Curr. Students. maths - marks.

↓ ↓ ↓

database Schema Table name.

name name name.

eg.2 ↓ ↓ ↓

Laptop School Results

SELECT * FROM database. Schema. Table - name.

It allows us to SELECT all the Columns in the Table.

SELECT DISTINCT → different values in that column.
Column - name.

FROM database. Schema. table - name;

Curr. Students. Maths - marks.

Student ID	name	Subject	Grade	Marks
101	Rhafi	Maths	10	99%
102	Mbali	Maths	10	20%
103	Fifi	Maths	11	50%

SELECT DISTINCT
FROM Curr. Students. Maths - marks;

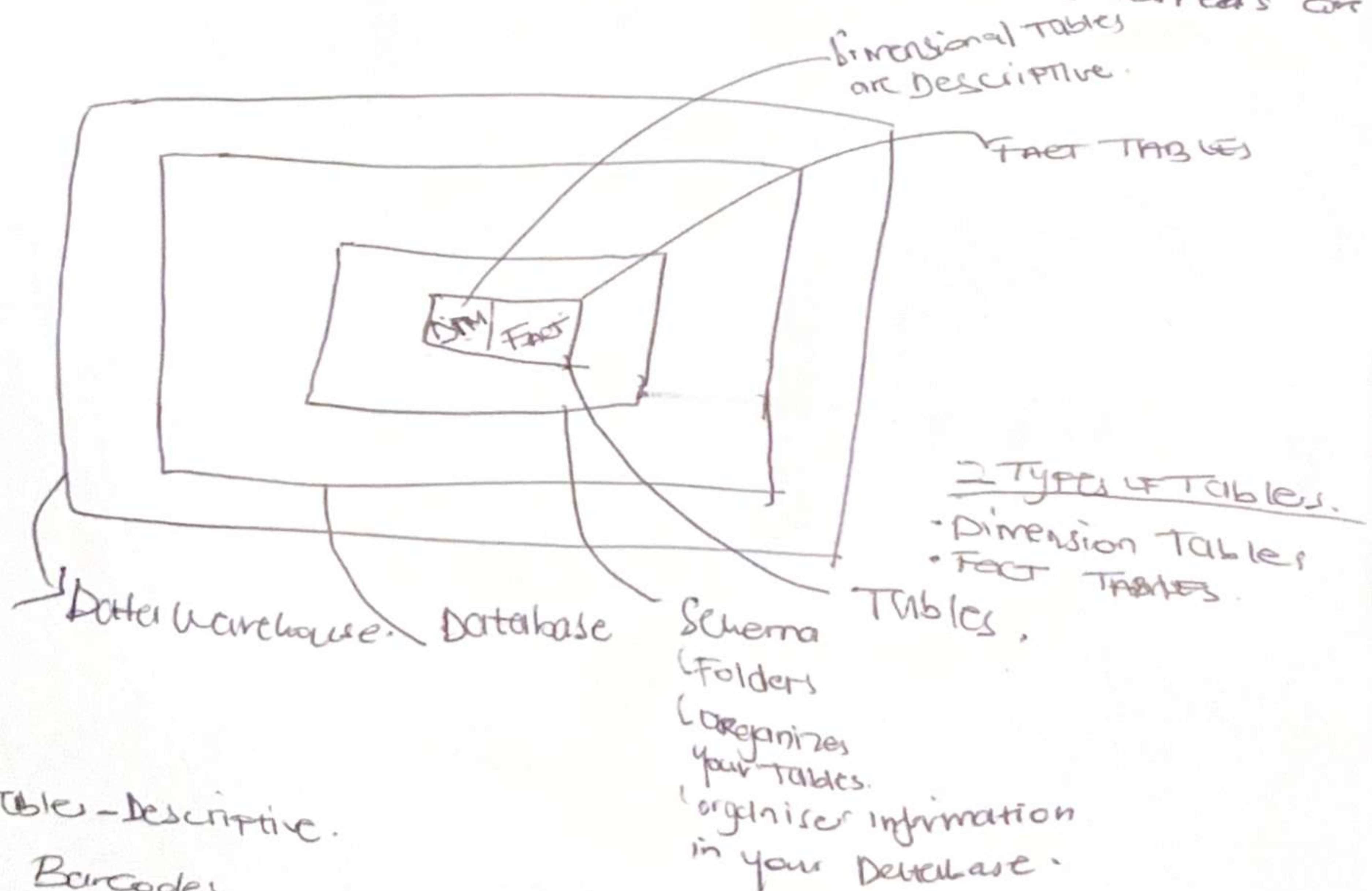
Day 4: DATA ANALYTICS CLASS, 2024.

INSTRUCTION: Google Bright-Learn CO. 29 → Play Around with APP
KATEKO - ADMIN → MUDITHANA ASSURANCE.



LAYERS IN THE DATA WAREHOUSE.

ENTER LAYER - Databases → inside the database we have Schemas.
we use Schemas to divide database. → Inside Schemas are TABLES.



Dimensional Table - Descriptive.

- O.S. Product Barcodes

Product ID	Product Name
1234	PROTEX
4321	COLGATE
3421	DETEL

FACT TABLE

ORDER ID	Product ID
101	

DIMENSIONAL TABLE



INFORMATION TABLE



FACT TABLE

Normally gives you information
about a specific product.

Fact Table



QUANTITATIVE (numbers)

Foreign key.

ORDER ID	PRODUCT ID	Quantity
101		

AGGREGATE FUNCTION. (RETURNS SINGLE VALUE)

Aggregate functions helps us to Consolidated, Summarises and combines together our data into groups.
Aggregate Function in SQL, Summarise many Rows into one Result per Group.

WE HAVE 5 Aggregate Functions:

- ① COUNT () → Counts How many Rows.
- ② SUM () → Sums up the data in a numeric Column.
- ③ AVG () → Gets average of a numeric Column.
- ④ MIN () → Looks for the Smallest Value in a Column.
- ⑤ MAX () → Looks for a Largest Value in a Column.

→ An aggregate function looks at multiple Rows in a Table and Return a Single Value.

Revenue
Apple
Banana
Orange
Kiwi

ID	Fruit name	Revenue
101	Apple	2
102	Banana	4
103	Orange	3
104	Kiwi	5

Aggregate.
 $2 + 4 + 3 + 5 = 14$.

SELECT SUM (Revenue) AS total - Revenue
FROM Products;

example ①

SELECT MAX (Revenue) AS highest Revenue This looks for the maximum Revenue in the Columns.
FROM Products;

Answer:

Highest - Revenue
5

example ②

SELECT MIN (Revenue) AS lowest Revenue
FROM Products;
= Minimum lowest - Rev
This looks for the Smallest or Minimum Value in the Column.

Minimum lowest - Rev
2

example ③

SELECT COUNT (ID) AS Number of Fruits.
FROM Products;

Number of Fruits
4

→ An aggregate function is used to Perform Calculation in the Rows of our table in a specific Column and Return a Single Value.

M - MALE

F - FEMALE

GRADE 10 H

M	M
F	M
M	F
F	M

= 5 males.

3 females.

used GROUP BY FUNCTION.

GROUP BY

ID	NAME	Gender
101	Methi	F
102	Rhota	M
103	Thabu	M
104	Jacob	M
105	John	M
106	Leotto	F
107	Robert	M
108	Ama.	F

WRITE AN SQL SCRIPT TO COUNT NUMBER OF Males and Number of Females.

SELECT COUNT (ID) AS number of students, gender
FROM STUDENTS GRADE 10 H.
GROUP BY Gender;

NUMBER OF STUDENT	Gender
5	M
3	F

BETWEEN STATEMENT.

- BETWEEN allows us to specify a range of values in a column that we want to filter from.
- Allows us to filter the data on a specific range of values in a given column.

- BETWEEN statement is inclusive.

SYNTAX.

```
SELECT Column1,  
        Column2  
FROM Table  
WHERE Column BETWEEN min and Max value;
```

SQL SCRIPTS.

```
SELECT Name,  
        Subject,  
        Marks  
FROM Curro.Student-Marks  
WHERE Marks BETWEEN 0 AND 30;
```

Name	SUBJECT	Marks
Mbali	Maths	22

SQL 2.

```
SELECT name,  
        Subject,  
        Marks  
FROM Curro.Student-Marks.  
WHERE Marks BETWEEN 50 AND 100;
```

Name	SUBJECT	Marks
Rhafi	Maths	99
Ierato	Geo	60
Niko	Science	70

NOT STATEMENT.

- NOT STATEMENT Gives you results contrary to what we're looking for.
- NOT OPERATOR Gives us the ability to exclude values in a column.

```
SELECT Column1,  
        Column2  
FROM database.Schema.Table  
WHERE NOT Condition
```

```
SELECT Column1,  
        Column2  
FROM database.Schema.Table  
WHERE Column NOT BETWEEN min  
and Max value.
```

```
SELECT Column1,  
        Column2  
FROM database.Schema.Table  
WHERE Marks NOT BETWEEN 50 AND 100;
```

Name	SUBJECT	Marks
Mbali	Maths	22
Amy	Geo	90

RECAP.

AND OPERATOR	OR OPERATOR
- Combines Multiple Conditions And Both Conditions must be met.	- Combines Multiple Conditions And at least 1 Condition must be met.

IN STATEMENT - Allows To Specify Multiple Values To Choose From When Filtering.

IN - is similar to the OR Statement Syntax.

SELECT COLUMN 1,
COLUMN 2

FROM TABLE
WHERE COLUMN IN (value1, value2, value3).

QUESTION Example.

Curro Students & Marks

Student ID	Name	Subject	Marks
101	John	Maths	99
102	Mbali	Maths	23
103	Lernato	Geo	50
104	Aly	Acc	40
105	Niko	Science	70

Give Names of Students doing Geo & Accounting.

Syntax

SELECT NAME,
SUBJECT

FROM Curro.Students.Marks.

WHERE SUBJECT IN ('Geo', 'Acc')

Result:

Name	SUBJECT
Lernato	Geo
Aly	Acc

SQL: NAME,
SUBJECT
FROM CURR. STUDENTS - MARKS
WHERE SUBJECT (NOT IN ('Geo', 'Art'))

Name	Subject
Rhofi	Maths
Mheli	Maths
Niko	Science

SALES

ID	Product	Category	Amounts
101	Pen	Stationery	10
102	Book	Stationery	15
103	Mobile	Electronics	100
104	Laptop	Electronics	200
105	Kettle	Appliances	90

WRITE AN SQL SCRIPT TO Calculate the Money Generated Per Category.

~~WRITE AN SQL~~

SELECT CATEGORY

FROM SALES

SUM (AMOUNT) AS total amount

GROUP BY CATEGORY;

Correction,

SELECT CATEGORY,

SUM (AMOUNT) AS Revenue

FROM SALES

GROUP BY CATEGORY;

CATEGORY	Stationery
Stationery	25
ELECTRONICS	300
APPLIANCES	90

'HAVING SYNTAX'

SELECT AGG-FUN (column) AS New-Column,
Categorical Column

FROM TABLE

GROUP BY CATEGORICAL column.

HAVING AGG-FUN (column) condition.

HAVING COMES AFTER AGGREGATION.

You Provide Before you Have,